RAG Learning

1.1 Number: Arithmetic and Place Value

OCR GCSE Mathematics (J560) — Component 01 (Paper 1, Non-Calculator), Component 02 (Paper 2, Calculator), Component 03 (Paper 3, Calculator)

1.1.1 Place Value and Ordering

Every digit in a number has a **place value** — its worth depends on its position. Understanding place value is the foundation of all arithmetic.

The Place Value Chart

Millions	Hund.Th.	Ten Th.	Thousands	Hundreds	Tens	Units	Tenths (.)	Hundredths
1,000,000	100,000	10,000	1,000	100	10	1	1/10	1/100
3	4	7	2	9	0	5	. 6	8

The number 3,472,905.68 shown in a place value chart. The digit 5 is in the units column, so its value is 5.

In the number **3,472,905.68**:

- The digit **3** has a value of 3,000,000 (three million)
- The digit **9** has a value of 900 (nine hundred)
- The digit **5** has a value of 5 (five units)
- The digit **6** has a value of 0.6 (six tenths)
- The digit **8** has a value of 0.08 (eight hundredths)

Ordering Numbers

To order a list of numbers — integers or decimals — compare digits column by column from left to right. Always align decimal points before comparing decimals.

Worked Example:

Place in ascending order: 4.3, 4.09, 4.31, 4.8, 4.09

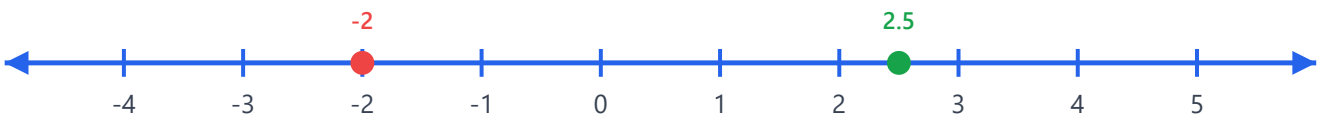
Align decimals:
4.30
4.09
4.31
4.80

Comparing the tenths column: 0 < 3 < 8

Within the 4.3x group, 4.30 < 4.31

Answer: 4.09, 4.30, 4.31, 4.80 → 4.09, 4.3, 4.31, 4.8

The Number Line



A number line from -4 to 5. Numbers increase to the right. -2 (red) and 2.5 (green) are marked.

Exam Tip: When ordering decimals, do not assume more digits means a bigger number. 4.9 is bigger than 4.81 because the tenths digit (9 > 8) comes first.

1.1.2 The Four Operations

The four operations are **addition (+)**, **subtraction (−)**, **multiplication (×)** and **division (÷)**. On Paper 1 (non-calculator) you must be able to apply these confidently to integers and decimals using written methods.

Addition and Subtraction of Decimals

Always align the decimal points in a column calculation. Insert zeros as placeholders where needed.

Worked Example:

Calculate $34.7 + 8.45$

```
  34.70
+  8.45
------
  43.15
```

Align decimal points; 34.7 becomes 34.70. Add column by column right to left, carrying as needed.

Answer: 43.15

Worked Example:

Calculate $102.3 - 47.85$

```
  102.30
-  47.85
--------
   54.45
```

Insert a zero so both numbers have the same decimal places, then subtract using decomposition (borrowing).

Answer: 54.45

Multiplication of Integers

The standard **long multiplication** method works for any integers. Multiply by each digit of the second number in turn, shifting one place left each row.

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Worked Example:

Calculate 324×47

```
    324
  ×  47
  -----
   2268   (324 × 7)
 + 12960   (324 × 40 — shift one place left)
  -----
  15228
```

Answer: 15,228

Multiplication of Decimals

Ignore decimal points, multiply as integers, then count the total number of decimal places in both original numbers and insert the point in the answer.

Worked Example:

Calculate 3.6×0.04

$36 \times 4 = 144$

Total decimal places: 1 (in 3.6) + 2 (in 0.04) = 3

Answer: 0.144

Division — Short and Long Division

For **short division**, divide by a single-digit number carrying remainders mentally. For **long division**, write out each step when dividing by a two-or-more digit number.

Worked Example:

Calculate $856 \div 8$ using short division

```
      107
   --------
8 | 856
      8
      --
       05
        0
        --
         56
         56
         --
          0
```

Answer: 107

Worked Example:

Calculate $1547 \div 23$ using long division

```
        67
     ------
23 | 1547
      138
      ---
       167
       161
       ---
         6
```

$23 \times 6 = 138$; remainder $1547 - 1380 = 167$; $23 \times 7 = 161$; remainder 6.

Answer: 67 remainder 6, or as a decimal: $67\frac{6}{23} \approx 67.26$

Exam Tip: When dividing decimals, convert to an integer division first. For example, $4.8 \div 0.06 = 480 \div 6 = 80$ (multiply both by 100).

1.1.3 Order of Operations (BIDMAS / BODMAS)

BIDMAS (or BODMAS) is the order in which operations must be carried out:
Brackets → **I**ndices (powers/roots) → **D**ivision → **M**ultiplication → **A**ddition → **S**ubtraction.
Division and Multiplication have equal priority (left to right); Addition and Subtraction have equal priority (left to right).

Step	Operation	Example
1st	Brackets	$(2+3)\times 4 = 5\times 4 = 20$
2nd	Indices (powers & roots)	$3+2^3 = 3+8 = 11$
3rd	Division & Multiplication (L to R)	$12\div 3\times 2 = 4\times 2 = 8$
4th	Addition & Subtraction (L to R)	$10-3+2 = 7+2 = 9$

Worked Example:
Evaluate $5+3\times(8-2)^2\div 9$

Step 1 — Brackets: $8-2=6 \;\rightarrow\; 5+3\times 6^2\div 9$

Step 2 — Indices: $6^2=36 \;\rightarrow\; 5+3\times 36\div 9$

Step 3 — Multiply/Divide (left to right): $3\times 36=108$; then $108\div 9=12 \;\rightarrow\; 5+12$

Step 4 — Add: $5+12=17$

Answer: 17

Exam Tip: Division and Multiplication are carried out left to right — they do NOT have a fixed order relative to each other. $12\div 3\times 2=8$, not 2. The same applies to Addition and Subtraction.

Worked Example:

Insert brackets to make $3 + 4 \times 2 = 14$ true.

$(3 + 4) \times 2 = 7 \times 2 = 14$ ✓

Answer: $(3 + 4) \times 2$

1.1.4 Negative Numbers

Negative numbers are numbers less than zero. They appear to the left of zero on the number line. Key skills include ordering, adding, subtracting, multiplying and dividing with negative numbers.

Adding and Subtracting Negative Numbers

- Adding a negative is the same as subtracting: $5 + (-3) = 5 - 3 = 2$
- Subtracting a negative is the same as adding: $5 - (-3) = 5 + 3 = 8$
- Two signs next to each other: same signs give **+**, different signs give **−**

Expression	Equivalent	Result
$3 + (-4)$	$3 - 4$	-1
$-5 + (-2)$	$-5 - 2$	-7
$7 - (-3)$	$7 + 3$	10
$-4 - (-6)$	$-4 + 6$	2

Multiplying and Dividing Negative Numbers

- **Positive × Positive = Positive** $(+) \times (+) = (+)$

- **Negative × Negative = Positive** $(-) \times (-) = (+)$
- **Positive × Negative = Negative** $(+) \times (-) = (-)$
- **Negative × Positive = Negative** $(-) \times (+) = (-)$

The same rules apply for division.

Worked Example:
Calculate: (a) $(-4) \times (-7)$ (b) $(-36) \div 9$ (c) $(-3)^2$ (d) $(-2)^3$

(a) $(-4) \times (-7) = +28$ (negative × negative = positive)

(b) $(-36) \div 9 = -4$ (negative ÷ positive = negative)

(c) $(-3)^2 = (-3) \times (-3) = +9$

(d) $(-2)^3 = (-2) \times (-2) \times (-2) = 4 \times (-2) = -8$

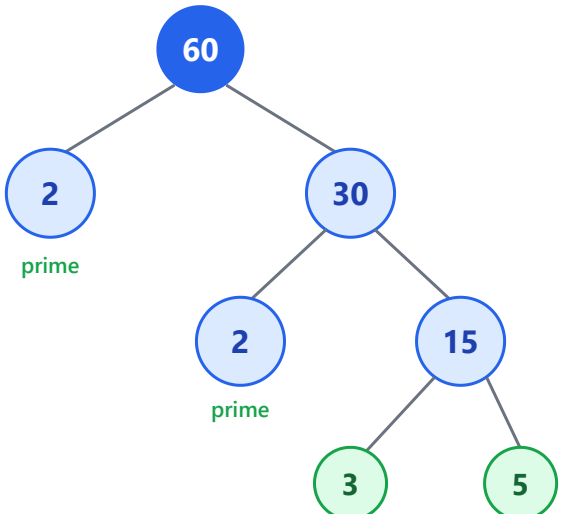
Exam Tip: $(-3)^2 = 9$ but $-3^2 = -9$. The absence of brackets means only 3 is squared; the negative sign is applied afterwards. This is a very common source of errors.

1.1.5 Multiples, Factors and Primes

- **Multiple:** a number in the times table of another. Multiples of 6: 6, 12, 18, 24, 30, ...
- **Factor:** a number that divides exactly into another. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24
- **Prime number:** a number with exactly two factors: 1 and itself. The first primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, ...
- **1 is NOT a prime number** (it has only one factor).
- **2 is the only even prime number.**
- **Composite number:** a positive integer with more than two factors (not prime and not 1).

Prime Factorisation (Factor Trees)

Every integer greater than 1 can be written as a unique product of prime factors. We use a **factor tree** to find these.



Factor tree for 60. The prime factors are 2, 2, 3, 5 (green circles), so $60 = 2^2 \times 3 \times 5$.

Worked Example:

Write 360 as a product of its prime factors.

$360 = 2 \times 180 = 2 \times 2 \times 90 = 2 \times 2 \times 2 \times 45 = 2 \times 2 \times 2 \times 9 \times 5 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$

$$360 = 2^3 \times 3^2 \times 5$$

Highest Common Factor (HCF)

The **Highest Common Factor (HCF)** of two numbers is the largest number that is a factor of both. To find the HCF using prime factorisation: find the prime factors of each number and multiply together the primes they share (using the lower power if they appear more than once in either).

Worked Example:

Find the HCF of 36 and 60.

$36 = 2^2 \times 3^2$

$60 = 2^2 \times 3 \times 5$

Common prime factors: 2^2 and 3^1 (take the lower power of each)

$$\text{HCF} = 2^2 \times 3 = 4 \times 3 = 12$$

Lowest Common Multiple (LCM)

The **Lowest Common Multiple (LCM)** of two numbers is the smallest number that is a multiple of both. To find the LCM using prime factorisation: multiply together the prime factors of each number using the *higher* power of any shared primes.

Worked Example:
Find the LCM of 36 and 60.

$36 = 2^2 \times 3^2$

$60 = 2^2 \times 3 \times 5$

Take the highest power of each prime: 2^2, 3^2, 5^1

$$\text{LCM} = 2^2 \times 3^2 \times 5 = 4 \times 9 \times 5 = 180$$

Exam Tip: A helpful check: $\text{HCF} \times \text{LCM} = a \times b$ (the product of the two original numbers). For 36 and 60: $12 \times 180 = 2160 = 36 \times 60$ ✓

Using HCF and LCM in Context

- **HCF is used for:** cutting/splitting things into equal groups, simplifying fractions, finding the largest tile that fits a floor.
- **LCM is used for:** finding when two recurring events coincide (e.g. buses), adding fractions with different denominators.

Worked Example:

Bus A runs every 12 minutes. Bus B runs every 20 minutes. They both leave at 9:00 am. When do they next leave at the same time?

Find LCM(12, 20): $12 = 2^2 \times 3$; $20 = 2^2 \times 5$

LCM $= 2^2 \times 3 \times 5 = 60$ minutes

Answer: 10:00 am

Exam Tip: Always express your final prime factorisation using index notation (e.g. $2^3 \times 3^2$). OCR mark schemes often require this form.